Swiggy Restaurant Recommendation System

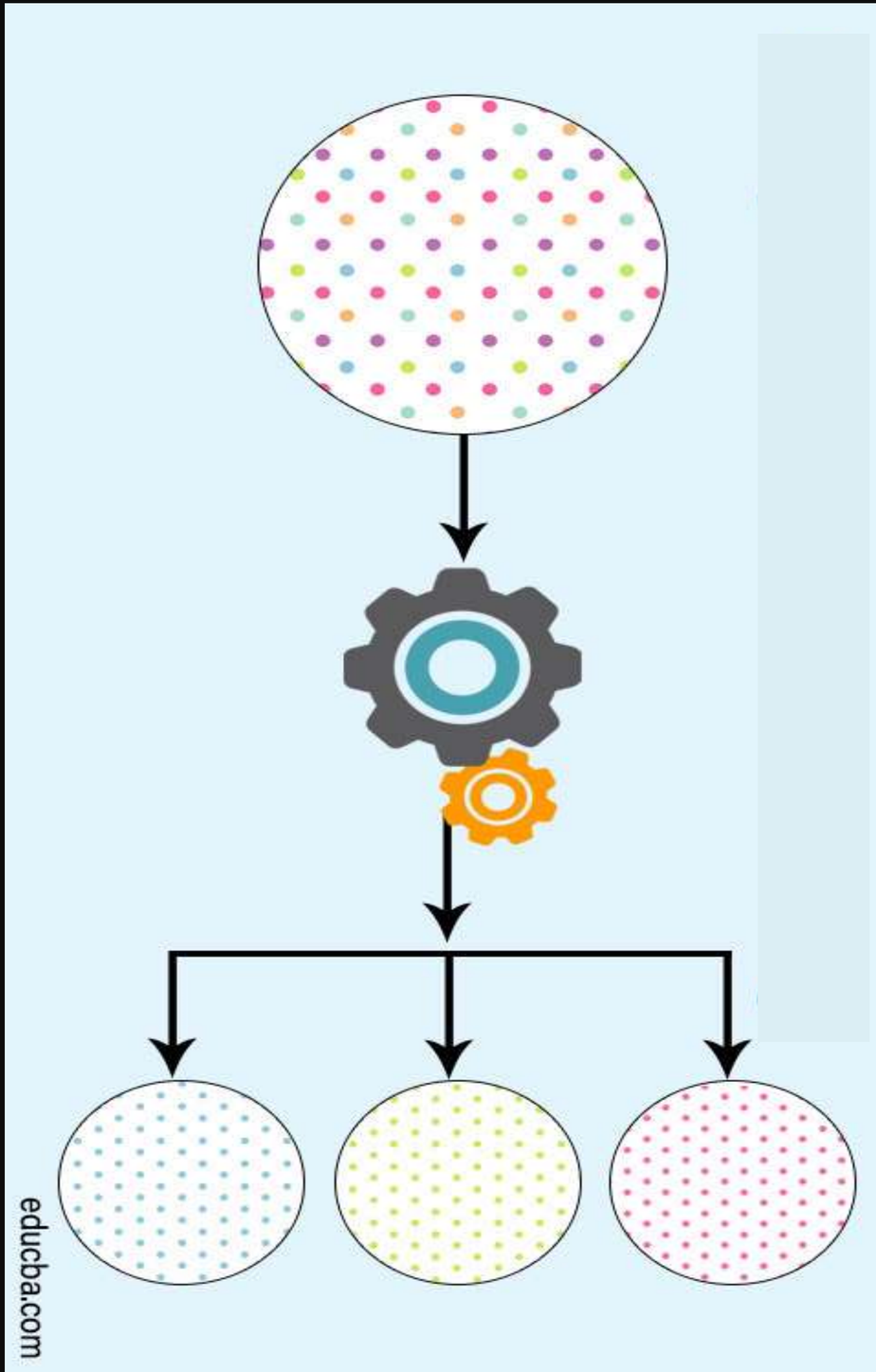
INTRODUCTION

1. This project develops a **Restaurant Recommendation System inspired by Swiggy** to help users easily find suitable restaurants.

2. The system recommends restaurants based on **user preferences such as city, cuisine, rating, and cost**.

3. It uses **data preprocessing, One-Hot Encoding, and K-Means clustering** to analyze restaurant data.

4. An interactive **Stream lit dashboard** allows users to input preferences and view recommended restaurants in real time.



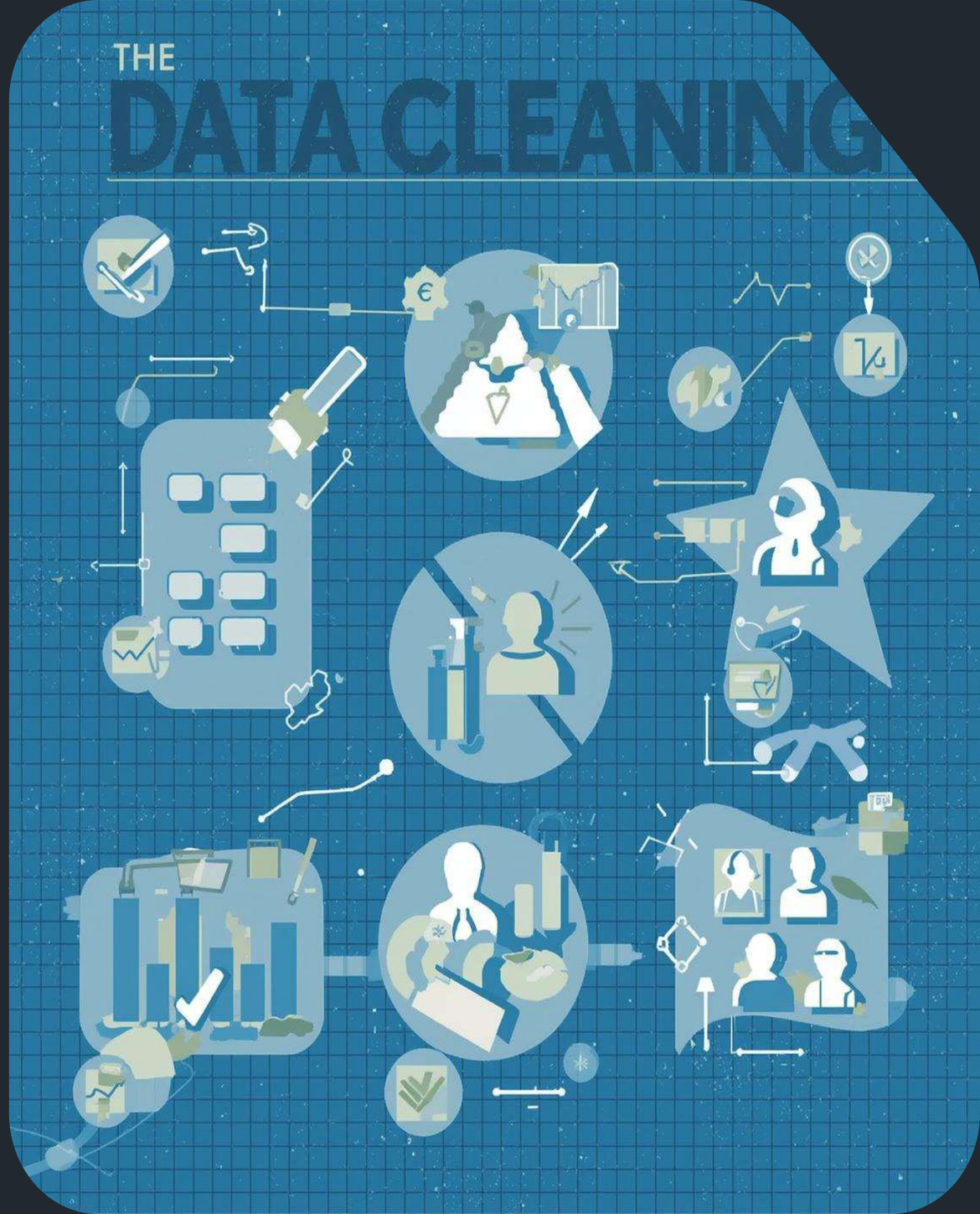
TOOLS USED

1. **Python** : It is used as the main programming language to build the recommendation system and perform data analysis.

2. **Pandas: It** is used for data cleaning, preprocessing, and handling the restaurant dataset.

3. **Scikit-learn: It** is used to apply **One-Hot Encoding and the K-Means clustering algorithm** for generating recommendations.

4. **Stream lit** : It is used to create an interactive dashboard where users can select preferences and view recommended restaurants.



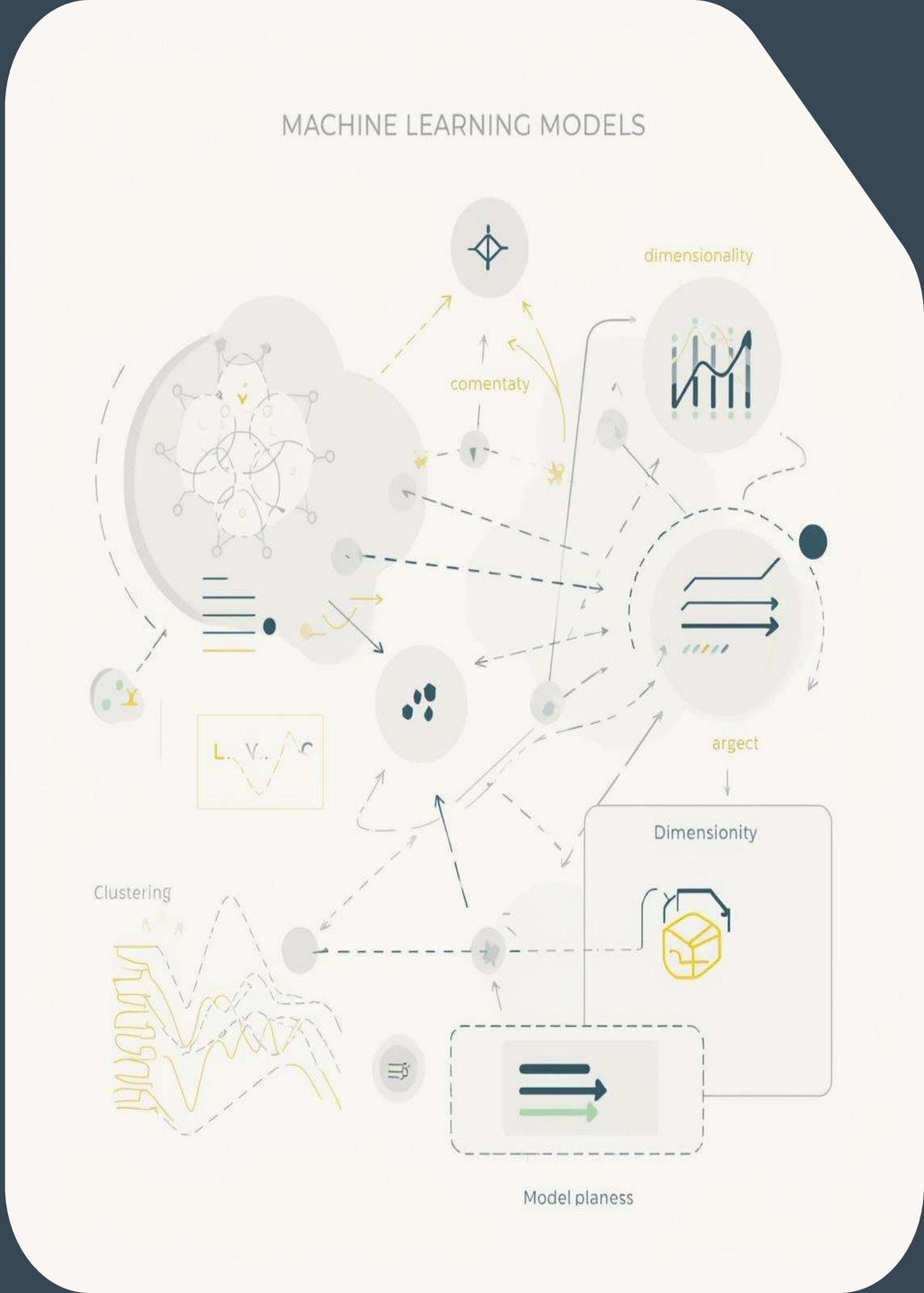
OBJECTIVE

1. Build a restaurant recommendation system.

2. Use data preprocessing and encoding techniques.

3. Apply clustering (K-Means) or similarity-based recommendation.

4. Develop an interactive dashboard using Stream lit.



SYSTEM ARCHITECTURE

USER INPUT

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DATA PREPROCESSING

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FEATURE ENCODING

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K MEANS CLUSTERING

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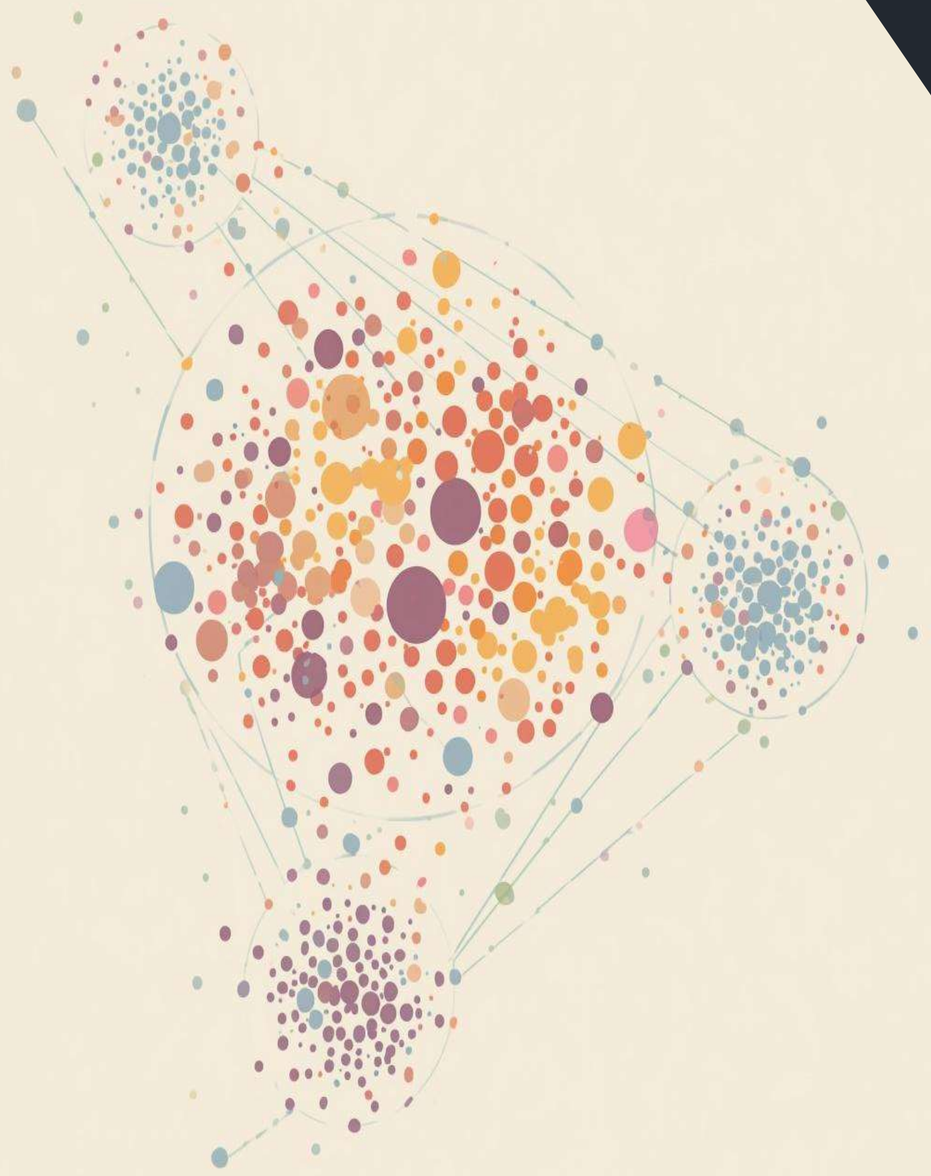
RESTAURANT RECOMMENDATION



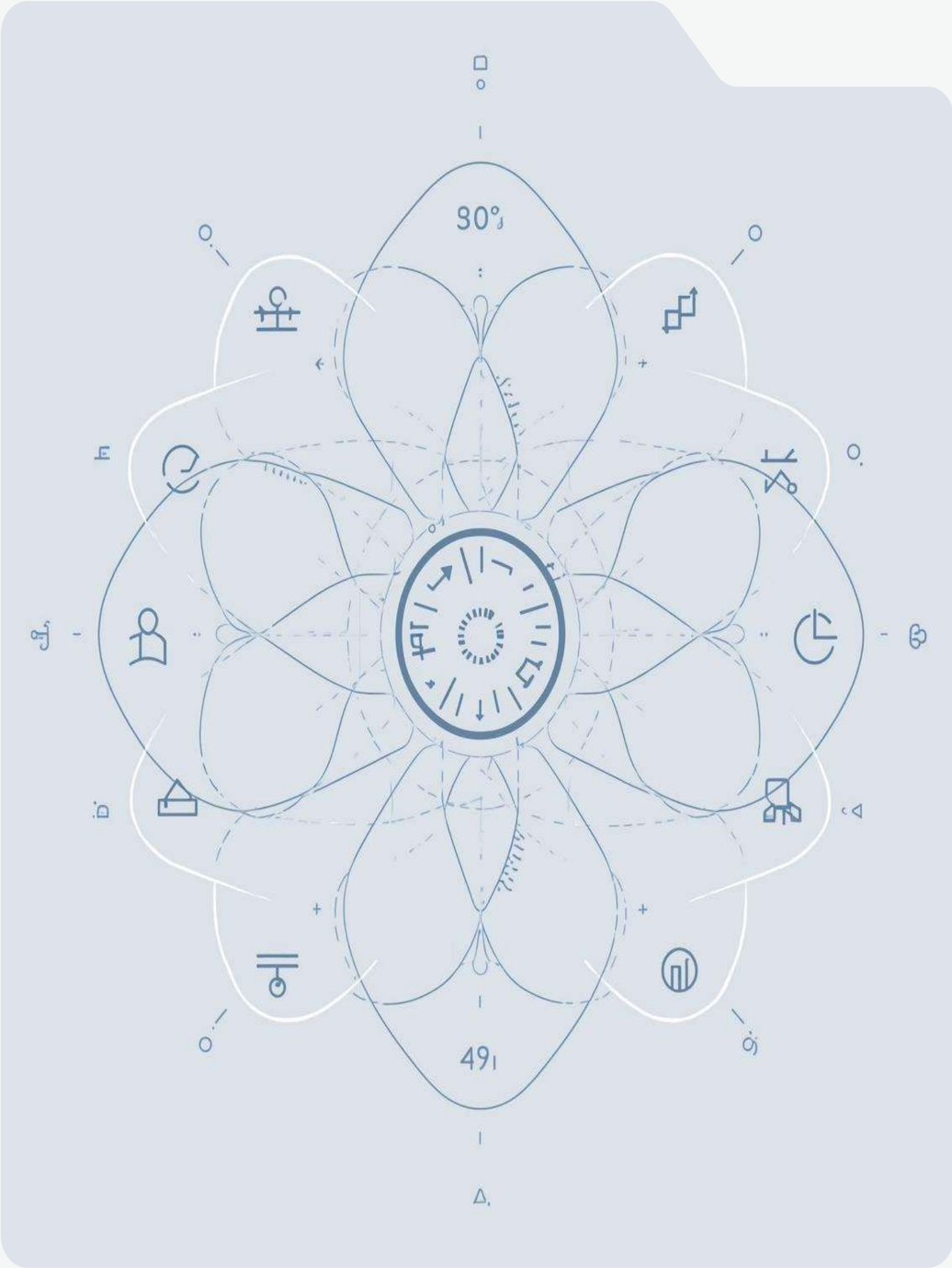
FUTURE SCOPE

1. The system can be improved by integrating **real-time restaurant data and APIs** from food delivery platforms.

2. Advanced recommendation techniques like **collaborative filtering and deep learning models** can be used for better personalization.

3. Additional features such as **user reviews, restaurant images, and location-based maps** can enhance the user experience.

4. The application can be **deployed on cloud platforms or mobile applications** to support large-scale users and real-world usage.



Thankyou

We welcome your feedback and question

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